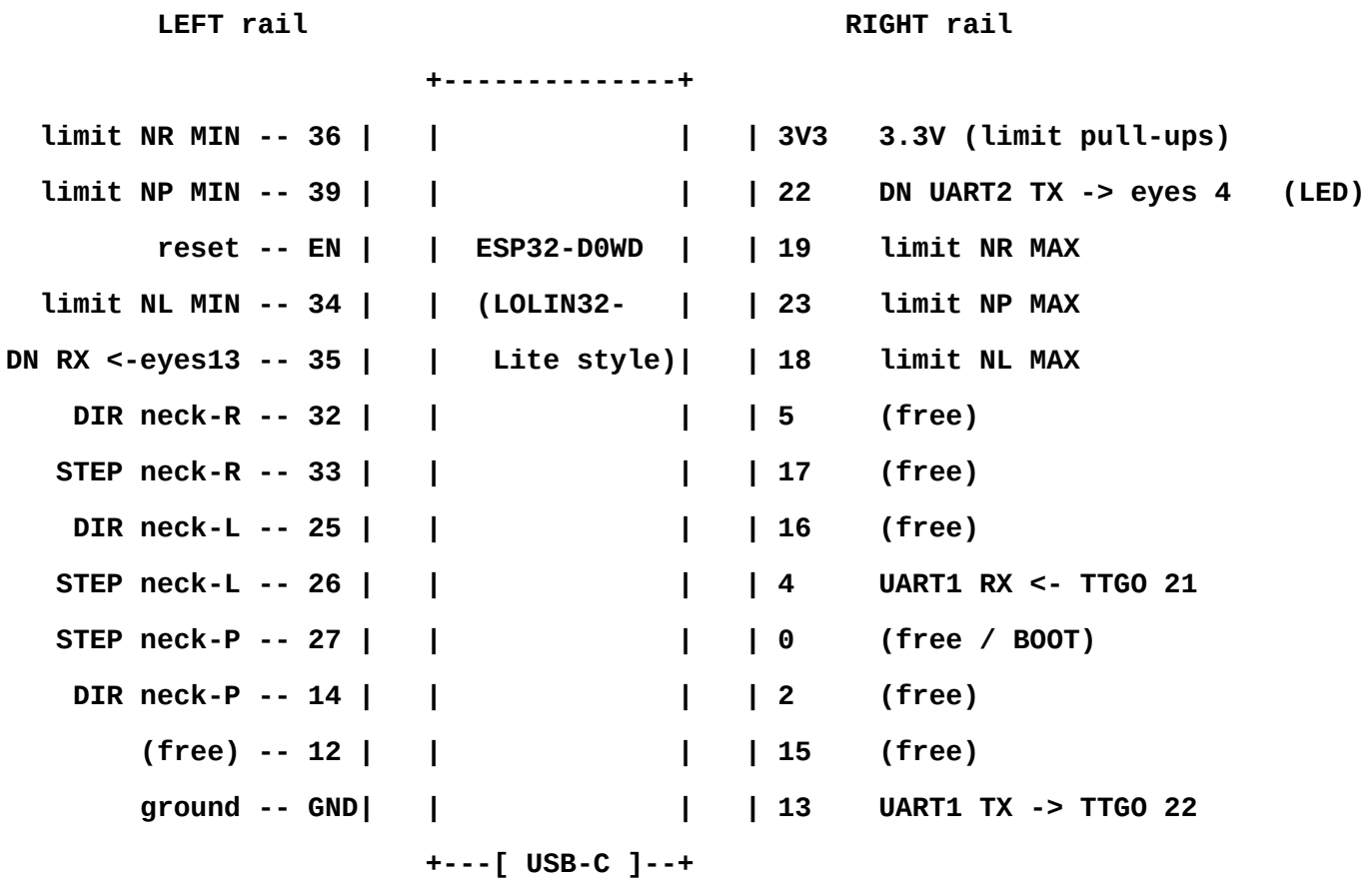
j4_stepper_neck -- pin diagram

ESP32-D0WD (LOLIN32-Lite style), component side up, USB-C at the BOTTOM



GPIO 21 is NOT present on this board, which is why the downstream TX is on 22.
Neck STEP/DIR pins feed the DM556TE PUL+/DIR+ inputs (PUL-/DIR- to GND).
Limits: one NC switch per endpoint. MIN ends (34/36/39) need external 10k
pull-ups to 3.3V; MAX ends (18/19/23) use the internal pull-ups.

Pin(s)	Function
26 / 25	neck-left STEP / DIR -> DM556TE PUL+ / DIR+
33 / 32	neck-right STEP / DIR -> DM556TE PUL+ / DIR+
27 / 14	neck-pivot STEP / DIR -> DM556TE PUL+ / DIR+
4 / 13	Upstream UART1 RX / TX -- RX from TTGO 21, TX to TTGO 22
22 / 35	Downstream UART2 TX / RX -- TX to eyes 4, RX from eyes 13
34	neck-left MIN limit (input-only, external 10k pull-up)
36	neck-right MIN limit (input-only, external 10k pull-up)
39	neck-pivot MIN limit (input-only, external 10k pull-up)
18	neck-left MAX limit (internal pull-up)
19	neck-right MAX limit (internal pull-up)
23	neck-pivot MAX limit (internal pull-up)